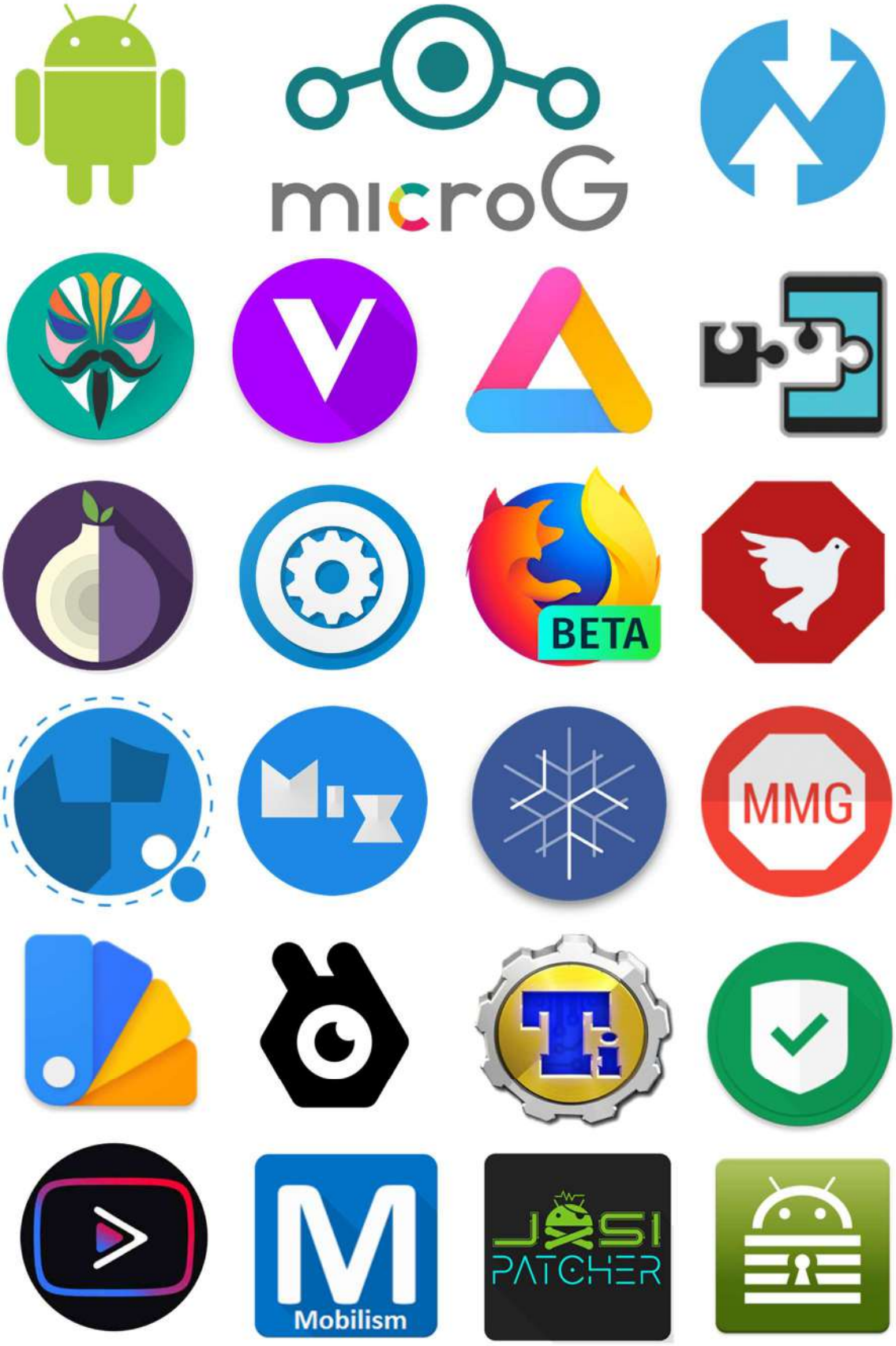
microG
BETA
MMG
Mobilism
JOSI
PATCHER

# [Guide] Build your own Ultimate Android Setup:

Did a new custom android build and damn it's better than the last one! :D

Let me sum up some of the sweetness:

- Data-Privacy! (All traffic goes through Tor, send back fake data to trackers, ripping Google's spy-ware out of the system and more)
- Advertisements, hell no, that whole system is the reason why our privacy is being but-raped right now.
- Battery usage? Good, even better than ever before!
- Performance? Also very good.
- Dark-theming and ultimate customization possibilities? Check
- Much more, won't sum it all up here, but those are the core benefits.

With us having to stay inside anyways tag along for my extensive guide and build your own ultimate Android device. It will be rather long, but detailed with many tips. I should note that this would wipe your device and that I'm not to be held responsible if you brick your device. However, you will not need any prior knowledge and I'll try to help if possible.

This guide should cover all officially LineageOS supported devices. However, I am sure there will be tips for other modders as well. I will be doing my setting up on a OnePlus 3T + LineageOS 17.1 (Q).

Things might be a bit different for you before flashing LineageOS has been done though. If that is the case, I hope you will be able to follow with context or a few additional searches on the web. This guide can also still be resourceful if you are on another custom ROM if you want to learn about more modules/tweaks.

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## 1. Info on the Android system:

The Android drive exists out of a few partitions:

- Boot (The code that boots up your operating system)
- System (Where your operating system resides)
- Recovery (Where you can go to recover your phone if system is f*ck up)
- Data (Where your personal data, pictures, applications, etc. are stored)

Android can boot into 3 different modes:

- System (The one you are familiar with)
- Recovery Mode (Where u can recover or reset your device)
- Fastboot Mode (A tool/protocol mode where you can send commands to your phone from your computer)

CMD (Command Prompt): Has nothing to do with Android but we will be using it, this is the default terminal from Windows, here you can enter CMD commands. To easily open a CMD window in a certain folder you can type 'cmd' in the path that is displayed near the top of a file explorer window that has that folder opened.

## 2. Unlocking the bootloader:

We will be unlocking the boot partition so we can modify it later to acquire system-less root access. To do this do the following on your phone to enable the developer options:

Settings -> About Phone -> Tap Build Number 7-8 times

Enter the developer options:

Settings -> System -> Advanced -> Developer options

Enable/Disable the following:

**Advanced restart** (**ON**, so you can easily reboot to the 3 different modes, otherwise you will have to look up the button combination you have to press for your device)

**OEM unlocking** (**ON**, with this we have the permission to unlock the bootloader in fastboot mode)

**Automatic system updates** (**OFF**, the modules tweaks we'll install later on can brick your system image if you don't disable them before doing a system update! You can safely re-enable them afterwards)

**Android debugging** (**ON**, so we are able to send ADB Android Debugging Bridge commands to our phone from our PC)

Now we will need to install the Minimal ABD and Fastboot tool on our computer. If you are an Android Developer and have the whole Android Studio installed then that should also suffice. I will be using Windows but I am sure there are other similar tools for Linux and Mac available.

Download and install from here, instructions on installation are also on the page: XDA Developers - [TOOL] Minimal ADB and Fastboot [2-9-18]

Connect your phone to a computer and reboot into Fastboot mode using one of following methods:

- If you enabled Advanced restart in developer options: Hold the power button -> Restart -> Bootloader
- Look online for the key combination for your device to boot into Fastboot mode, turn the phone off and hold down the key combination till it turns on
- Connect your phone to your PC and check your phone, you should get message where you need to confirm the usage of ADB
    - On your PC open a CMD window in the folder of the Minimal ADB and Fastboot tools and enter the following commands
    - adb devices (if all is okay your device should appear as a number as list of devices attached)
    - adb reboot bootloader (this will boot into Fastboot mode)

When you successfully rebooted into Fastboot mode enter the following commands in CMD to unlock your bootloader:

- fastboot devices (if all is okay your device should appear as a number as list of devices attached)
- fastboot oem unlock (this will unlock the bootloader but also fully wipe/factory reset the phone)
    - if that didn't work you can try the newer command to unlock the bootloader:
    - fastboot flashing unlock

Now I suggest you don't immediately reboot to system and stay in Fastboot mode for a bit. Since we also need to be in Fastboot mode for the next step.

## 3. Flashing a Custom Recovery (TWRP)

Stock recovery is limited in what it allows you to do (Factory reset etc).

That is why we'll be flashing a custom recovery instead, so we can unleash it from its chains, allowing us to do way more:

- Flash .ZIP modules and Custom ROMs
- Backup / Restore / Wipe partitions
- ADB Terminal
- …

I'll be showing you how to flash my personal favorite, TWRP (Team Win Recovery Project), it has a nice GUI and never failed me. Another honorable mention would be CWM (ClockWorkMod).

On your computer, look for the latest version for your device and download it from here: https://twrp.me/Devices/

If you found your device use the Download links with the mirrors like 'Primary Europe', pick the one closest to you. Then choose the latest .img available.

Place the downloaded .img file in the folder where the Minimal ABD and Fastboot tool is installed. And rename it to 'recovery.img' (The rename is optional but will make the typing of the commands easier).

Then enter a CMD window in that folder and type the following to TEST if the recovery you downloaded works for your device, for this your phone should still be in Fastboot mode, but I assume that it still is from unlocking the bootloader:

- fastboot devices (if all is okay your device should appear as a number as list of devices attached)
- fastboot boot recovery.img (Or the original name if you didn't rename)

This will just boot you into TWRP without doing the final flashing yet. So if you were to download a corrupt/wrong .img file you won't be stuck with a corrupted recovery.

If booting into TWRP was successful then it is safe to flash it so it's there to stay.

Boot back into Fastboot mode from TWRP by clicking Reboot->Bootloader

Then on the CMD window on your computer type the following commands:

- fastboot devices (if all is okay your device should appear as a number as list of devices attached)
- fastboot flash recovery recovery.img (Or the original name if you didn't rename)

We now successfully flashed TWRP. Again I suggest that you stay in TWRP for now as we'll be needing it in the next step.

## 4. Flashing LineageOS 17.1

LineageOS: Is an open source custom rom. The successor of the well-known CyanogenMod bringing you the improved version of stock Android but still the stock feeling. Extended with Privacy Guard (an in depth app permission-management system), without carrier spy/bloat-ware.

In my opinion this is the best custom rom to have these days.

I should also note that I absolutely recommend to go for a clean install! At first I migrated my normal LineageOS that had GAPPs which caused me various issues. But a clean install works perfectly.

On your computer, look for the latest .zip version for your device and download it from here: https://wiki.lineageos.org/devices/

Back in TWRP on your phone select:

Wipe->Advanced wipe->Select all the partitions->Swipe to Wipe

This will wipe the whole phone, preparing it for the clean installation for the custom rom.

Then transfer the downloaded lineageos.zip file from your computer to your phone.

Back in TWRP on your phone select:

Install->Install Zip->Navigate to the lineageos.zip file & click it->Swipe to confirm Flash

Then wait for the flashing to complete. After flashing you may select Wipe Cache/Dalvik. As soon as it's done your successfully installed LineageOS! This time it's not a suggestion, stay in TWRP until we have completed the next step! Otherwise, if you attempt to do the next step later, you might have a boot loop on your hands.

## 5. Flashing Magisk

If you remember the older days of android modding. We used to give applications root access by using ChainFire's SuperSU. And then modified our android by making changes to the system partition.

The big downside to this was that you couldn't receive OTA's (Over The Air updates) anymore. When you did do an update, it would mean losing all the changes you made to the system partition, needing to re-do them after each update. However, things have changed, for the better, and our trusted SuperSU is outdated now.

Nowadays we are going to be using the Magisk framework to handle our root access. And to handle our system changes, in the form of Magisk Modules.

The huge benefit to this is that Magisk doesn't alter the system partition. It merely places links in the system to the modules that contain the system changes. Meaning that our system partition will remain unaltered, and **we will be able to install OTA's (But before you do that make sure to disable all Magisk modules and reboot your system! After the system update it's safe to re-enable them again).**

On your computer, look for the latest Magisk-vXX.X.zip and the latest MagiskManager-vX.X.X.apk for your device and download them from here:

https://github.com/topjohnwu/Magisk/releases

Then transfer the downloaded .zip and .apk file from your computer to your phone.

Back in TWRP on your phone select:

Install->Navigate to the Magisk-vXX.X.zip file & click it->Swipe to confirm Flash

When the installation is done (unless you already wiped these after installing Magisk):

- Wipe->Advanced wipe->Select the Dalvik and the cache partitions->Swipe to Wipe
- Reboot->System

Yes finally we'll be booting into our fresh LineageOS! It will go through it's initial setup (Connecting to WiFi, registering fingerprint, etc). Fill this in to your contents.

Once you booted into your system:

Open up the file explorer app -> Navigate to where you saved the MagiskManager-vX.X.X.apk -> Click it and install it (If your device requests you to enable the installation of unknown sources, then enable it)

If you made it this far you should now be running under an unlocked bootloader with TWRP as your custom recovery and LineageOS as your custom ROM extended with the Magisk framework, nice!

## 6. Setting up LineageOS 17.1

Now that we finally booted to system we still have some necessary things to configure in LineageOS.

Enable the developer options:

Settings -> About Phone -> Tap Build Number 7-8 times

Enter the developer options:

- Settings -> System -> Advanced -> Developer options
- Set the following settings:
    - Advanced Restart (Enabled)

- OEM unlocking (Enabled)
- Automatic system updates (Disabled)
- Update recovery (Disabled)
- Window animation scale (.5x)
- Transition animation scale (.5x)
- Animator duration scale (.5x, these last 3 will cut the length of your animation speed in half, making your phone feel snappier)

## 7. Magisk Manager for Recovery:

This thing is awesome, when experimenting with modules you always have a chance that you'll end up with a bootloop. This module will save your ass if that's the case! Providing a safe and easy debugging process.

Originally created by VR25 and updated by myself to work with the latest V20 version of Magisk.

Download the MagiskManagerForRecovery_vX_XXXXXXXXX.zip file somewhere on your phone file from here and save it somewhere safe on your phone: https://github.com/Rikj000/mm/releases

You can flash this module in TWRP to disable magisk modules trough the terminal of your recovery in an easy interactive way.

Usuage:

- Boot into TWRP
- Install the MagiskManagerForRecovery_vX_XXXXXXXXX.zip
- Go to Advanced->Terminal window in TWRP & run the 'mm' command to start managing your modules
- Follow the instructions/wizard. Everything is interactive.

For now I suggest you download it and store it somewhere safely on your device.
If you ever get a bootloop due to a module you just boot into TWRP, open a terminal, type mm and you should easily be able too disable the faulty module. You will have to reflash it each time you want to use it but it will show up in the Magisk Manager as installed after the 1st time. Also since EdXposed framework is a systemless Magisk Module nowadays we can also use it for those modules, since you can disable 'riru_edxposed' in it, thus disabling the whole EdXposed framework, which will allow you to reboot and uninstall the faulty EdXposed module/application.

Also before attempting this, or when still unsuccessful after disabling a module it's also worthwhile to wipe Dalvik / ART Cache & Cache in TWRP before attempting your reboot. This can also often resolve a bootloop. You can do it by:

- Boot into TWRP
- Select Wipe->Advanced
- Select Dalvik / ART Cache and also Cache (2 toggles)
- Tap Wipe & Reboot after completion.

## 8. Magisk Modules:

Now that we have this awesome framework available let's start using it by adding some modules.

These didn't fit in a category of their own, you can just grab them from the Magisk Manager unless stated otherwise. Don't forget to reboot after installation to apply em.

### 1. Ainur Narsil / Sauron

System wide audio modification library that helps keep the playback track clean while using FX'es (aka equalizers)

### 2. Audio Modification Library

Framework that allows seamless integration of multiple Audio mods.

### 3. BusyBox for Android NDK

More of a dependency for other root applications. It allows for extension of their capabilities.

### 4. CloudFlareDNS4Magisk

CloudFlare provides a free, fast DNS provider. It will increase the speed of your internet a bit and it respects your privacy more than darn Google does. Also install the IPv6 variant.

### 5. GPU Turbo Boost

This module aims to increase Android powered devices by up to 75% but at the same time reduces power consumption by up to 25% seamlessly.

This one you have too download from GitHub and flash the zip in Magisk Manager:
https://github.com/EmperorEye1993/GPU-Turbo-Boost/releases

### 6. LiboemCrypto Disabler

Netix put some DRM protection on their content to block rooted users, this disables the library that does that, thus giving us back Netflix and all others that use this kind of DRM.

### 7. Universal GMS Doze

I'm unsure if this helps on MicroG devices since it's inferior, but it for sure does on GApps devices and I still install it since I really don't want to support Google with a single bit of data anymore.

This application will put some chains on the GApps, making it so that they cannot farm as extensively by blocking unnecessary wakelocks.

### 8. Wifi Bonding (Qcom)

This module can seriously help increase your internet speed, but it's only for Qualcomm based devices.

## 9. Installing Xposed Framework:

The Xposed Framework was originally created by rovo89, but has been followed up by a systemless version called EdXposed Framework.

The framework allows you to install modules, kinda like you would in Magisk. But these work differently. They use hooks to catch variables and/or methods to manipulate application behavior. Allowing you to do some really cool/powerfull stuff.

- Download and install the following modules from within Magisk Manager:
  - Riru – Core (Dependency for EdXposed)
  - Riru – EdXposed (YAFHA or SandHook, you'll have to try to see what works for your device)
- Download and install the EdXposed Manager.apk from here: https://github.com/ElderDrivers/EdXposedManager/releases
- Open up EdXposed Manager and go to Settings (Framework section)
  - Enable App List Mode
  - Enable Force hook modules (previous step should automatically enable this by default)
- Reboot your device

Go to Modules, enable the EdXposed Manager module and reboot.

Open up the EdXposed Manager->Settings->Silent installation (root). This won't be requested by Magisk until you install your 1st module. There won't be a specific Xposed Modules section in the guide since each app/module will have it's own section.

## 10. Installing Aurora Store, Droid & Services

We're gonna want to be able to use our everyday apps but without the evilness from Google lurking in the back.

To be able to do that we'll be using Aurora Store; an open source, privacy respecting alternative to the Play Store. It connects with the same database as the Play Store, so you'll see the same apps, with the same versions, but will be more private while browsing for apps!

Aurora Droid is an alternative to F-Droid, our known and trusted app store for open source apps. This one uses the nice modern GUI that Aurora Store also uses thus I prefer it. However F-Droid has been around for a long time and is rather stable in combination with it's Privileged Extension, Aurora Droid is rather new and still has some bugs, but I'll teach you how to deal with them at the end of this section.

Aurora Services is an addon that we'll be using to make them both work nicely in the background.

Download Aurora Stora and Aurora Droid from here:

https://auroraoss.com/downloads.php

And Aurora Services can be downloaded from here, choose the Magisk .ZIP version:

https://gitlab.com/AuroraOSS/AuroraServices/-/releases

Install all 3 .APK files on your device.

Now we'll need to do some additional things to make em work nicely in the background:

- From within Aurora Store download & install 'Termux'
- Give Termux folder permissions
- From within Magisk Manager download & install 'App Systemizer (Terminal Emulator)'
- Reboot your device to enable the magisk module
- Open up Termux
- Type 'su' (give it root permission if Magisk asks)

- Type ‘systemize’
- Follow the on screen steps too systemize Aurora Store, Aurora Drive and Aurora Services (Add them ass app/priv, you should only systemize apps that really need it)
- Reboot your device to enable the newly Systemized apps

Now when App Systemizer is enabled in Magisk the systemized apps will be running as system privileged apps while being able to make em back normal by just toggling the module in Magisk and rebooting.

We still need to setup the Aurora Services, to do this:

- Open up Aurora Store
  - Go to Settings->Installations->Aurora Services
  - Give it the permissions it asks & Also Whitelist Aurora Store and Aurora Droid
  - Go to Settings->Installations->Installation Method: Set to Aurora Services
  - Go to Settings->Updates->Updates Interval-> Set to 1 day (or whatever you like)
- Open up Aurora Droid
  - Go to Settings->Installations->Installation Method: Set to Aurora Services
  - Go to Settings->Updates->Updates Interval-> Set to 1 day (or whatever you like)

With this both the Google Apps as the Open Source apps have a nice looking Appstore that respects your privacy and will auto update for you.

You can easily add more Repositories (Places where it looks for apps) in Aurora Droid with the Repositories List option.

For now it has some know bugs that hopefully get patched as it gets more popular but these are easy to work around by following these tips:

- Do not add Unavailable repo’s! They will cause the sync to hang until you force stop the app.
- If you can’t search in Aurora Droid sync the repositories
- If apps aren’t downloading: You can go to Downloads-> Triple dot icon -> Force Clear All
- If apps aren’t installing in Aurora Store: Settings-> Installations ->Force clear installation sessions
- On top of the 2 above you can go to the app’s info, force stop it and clear it’s cache
- If that doesn’t work reboot your device and it should work again
- Worst case scenario, open up MiXplorer (or any other file manager) at your internal storage’s root open the Aurora folder, there you should be able to find your apk and be able to install it manually, if it are multiple files you’ll need SAI (Split APKs Installer), covered later in this guide.

## 11. Installing MicroG

While now we’re able to get Play Store apps on our devices, most of them still won’t work (properly). This is because we didn’t install the GAPPs, the Google System apps. These are embedded in Android as system apps upon installation and they provide the Google Services, Google Play store, Google Cloud Messaging (meaning your notifications), Google Maps API etc.

MicroG: An open-source alternative to the GAPPs system apps. It replaces the with spy-ware filled closed source battery sucking Google stuff. With an open source lightweight alternative, that sends back the bare minimum to Google. Thus ripping out Googles claws out of your system / private life.

I find this to be the hardest part of the whole setup, but bear with me, I'll walk you through it.

Make sure you added the microG repository in

- Open Aurora Droid (add the microG Project repository & sync if you haven't yet)
- Go to Home->Repositories->microG
- Install the applications in the following order:
    1. microG Services Core
    2. microG Services Frqmework Proxy
    3. FakeStore
    4. microG DroidGuard Helper
- Go to Home->Repositories->F-Droid
- Install following applications:
    1. MozillaNlpBackend
    2. NominatimNlpBackend

Now we still need to spoof the signatures of the GAPPs so Google will think that we're legit. LineageOS doesn't support this out of the box for security reasons, but we'll be using a handy Xposed module that'll allow us to just spoof the signatures we need for GAPPs.

- Open EdXposed Manager
- Go to Download and look for "FakeGApps"
- Install it, enable it and reboot
- Open Aurora Store
- Look for "Google Play Services"
- Select Blacklist

Since we are spoofing the signature of Google Play Services we should blacklist it in Aurora Store, otherwise Aurora will think we have it installed and will try to install it over MicroG when their crap get's updated.

Normally the GApps would run as system apps, so just to be sure we will do the same, we really want these to be working all the time after all:

- Open up Termux
- Type 'su' (give it root permission if Magisk asks)
- Type 'systemize'
- Follow the on screen steps too systemize the following apps: (Add them ass app/priv, you should only systemize apps that really need it)
    - FakeGapps
    - FakeStore
    - microG DroidGuard Helper
    - microG Services Core
    - microG Services Framework Proxy
    - Mozilla UnifiedNlp Backend
    - Nominatim GeoCoder Backend
- Reboot your device to enable the changes

Alright now MicroG should be successfully installed but we still need to set it up.

Open up the microG Settings app

- Go to Self-Check->Permissions granted
  - Tap each unchecked box to bring up a popup to enable all needed permissions.
  - Or you could give it the permissions like you would for any other android app.
- Go to Self-Check->System
  - Tap Battery optimizations ignored to allow always running in the background.
- Add your google account
- Enable Google device registration
- Enable Google Cloud Messaging
- Enable Google SafetyNet
- UnifiedNIp Settings:
  - Configure location backends -> Enable Mozilla Location Service
  - Configure address lookup backends -> Enable Nominatim

Now there is 1 downside to this that I've noticed and that's that an alternative to Google's QR Code scanning library isn't implemented yet :/ So apps that depend on this won't be able to scan, however 9/10 they'll allow you to manually enter the code as well, we won't be doing this, since we can use an open source app to scan and copy the code to our clipboard so we can easily paste it where it's needed.

You can download the latest version of SecScanQR through Aurora Droid

Be happy, the hardest part is over by now! :)

## 12. Installing Orbot

Orbot will greatly enhance your anonymity by funneling all your traffic through the Tor network.

Beware, with this awesome addition comes a downside, it may decrease your internet speed from time too time, depending on through which nodes you are going at the moment. I found switching too a new Tor identity / turning it off/on again from within Orbot often helps with this though.

- Download and install it from Aurora Store
- Open it:
  - Enable VPN Mode and start the Orbot
  - Tap the cogwheel at VPN mode at the bottom and add each app (you'll have too manually add all apps here that you'll install in the future)
  - Go to it's settings:
    - Enable Start Orbot on Launch
    - Enable Allow Background Starts
- Open your android Settings:
  - Go to Network & internet->Advanced-> VPN->Orbot Settings
  - Enable Always-on VPN
  - Enable Block connections without VPN

For Android 10 users, after installing some more apps I had to disable the Block connections without VPN setting, since it was blocking some apps for me.

At VPN mode in the app I also disabled it for:

- Mobilism
- Vanced Manager

## 13. F*ck advertisements:

To completely get rid of advertisements we'll need to setup and install a handful of things.

### 1. AdAway

AdAway will be our main AdBlocker. It works by systemlessly editing your host file, where it blacklists all the ad and tracker domains. Meaning they won't come through and resulting in gray banners/areas where ads should have been + saving up your mobile data since the ad's won't come through.

- Open up Magisk Manager
- Open it and go to Settings
- Enable Systemless Hosts
- Download and install AdAway from Aurora Droid
- Open AdAway
- Go back to the "Home" section check for updates and apply them.
- Reboot your device after applying (come check for new hosts version from time to time + reboot to apply)

### 2. MinMinGuard

MinMinGuard will take care of those gray advertisement banners that AdAway leaves you with. It takes the height value of the ad-banners and changes it to 0px, rendering them invisible and giving you back the full screen view like some pro-mode would!

- Open up EdExposed Manager
- Download and enable MinMinGuard, reboot to apply

### 3. FireFox

I recommend FireFox as your browser since they target towards privacy + they have plugin support (kinda) and it has the ability to block trackers and does a pretty good job at it.

However the new v80 of FireFox won't allow you to install all the add-ons you want, due to their new BS add-on signing policy, from which you can't opt out for now.

So until they make this able to opt out, we'll hack them in the new version, since we don't deal with oppression on our devices.

Also we really need the OnionFlare Add-on, since otherwise we'll be greeted by CloudFlare's captcha hell on half of the websites. The official Tor Browser however is whitelisted and this add-on spoofs the fingerprint of it so CloudFlare will think we're using that, bypassing the captcha's.

- Download and install FireFox Beta from Aurora Store
- Open it up & go to Settings:
    - Search-> Set Default Search Engine to DuckDuckGo and deleta all others
    - Set as Default Browser

- Enhanced Tracking Protection->Set to Strict
    - Optionally but even better you could create your own Custom profile by following these guides:
        - https://www.privacytools.io/browsers/ g#webrtc
        - https://www.privacytools.io/browsers/#webrtc
- Data Collection->Disable all 3
- To hack in the add-ons follow this well-explained guide: https://www.reddit.com/r/firefox/comments/i8igam/androidrequires_root_install_any_addon_from_amo/
    - You'll have to replace org.mozilla.firefox with org.mozilla.firefox_beta when doing the adb commands
    - You can skip the creation of the collection if you are fine with using mine, it's filled with add-ons that block ads, enhance your privacy, Onionflare, Wayback Machine and Darkreader
        - Browse my collection
        - Download my collection's json file
- uBlock Origin will need some further configuration after installing all addons.
    - Open up it's settings->Filter lists->Explicitly enable all for following categories:
        - Built-In
        - Ads
        - Malware Domains
        - Multipurpose
        - Regions, languages (Don't do all here, only your native language)
    - Don't forget to press apply after adding all the filter lists

## 4. YouTube Vanced

YouTube Vanced is an awesome mod for YouTube, blocking ads, enabling background and picture-in-picture play and some more neat stuff.

Today that's pretty easy to do with the new Vanced Manager app that will handle the installation for you.

- Download it through FireFox from here: https://vanced.app/
- (You may have too temporarily disable Orbot and your VPN if the vanced version shows up unavailable, in the Vanced Manager this seems to be a bug for now.)
- Install the nonroot microG version
- Open up the Vanced MicroG settings from the Vanced Manager and set it up like you did for general MicroG.
- Install the nonroot Vanced version (We're going non root since we don't want to fully enable signature spoofing, that would be quite a security risk)
- Don't forget to ignored battery optimizations for Vanced MicroG to allow always running in the background. You'll have to do this from system settings.

### 5. Spotify

You can always trust on Balatan for a modded version of Spotify that will get those pesky ads out. Basically premium without saving locally.

- First register and login here: release-apk.com/
- Once logged in look for the latest .APK with this link: https://forum.release-apk.com/search.php?keywords=Spotify+%5BMod%5D&terms=all&author=Balatan&fid%5B%5D=60&sc=1&sf=titleonly&sr=posts&sk=t&sd=d&st=0&ch=300&t=0&submit=Search
  (Beware you want the [Mod] not the [Mod Lite] version)
- Install it & log in. I do so by email, I tend too not use Google/Facebook SSO logins since they give em even more data.
- Open up Aurora Store->Search for Spotify
- Tap the triple dots in the top right and add it to the blacklist, since we don't want Aurora Store to overwrite the modded version.

Now it should work! However you'll have too update this one manually from time to time so I suggest bookmarking the long URL above to your phone's Home Screen.

### 6. SoundCloud

Yes it is possible! But it requires a bit of tinkering as usual when you want nice stuff.

- Install the latest SoundCloud from Aurora Store
- Open up AdAway
- Go to "Your lists" and add: static.cdn.targetspot.com
- Go to "Hosts sources" and add:
  - https://filters.adtidy.org/extension/chromium/filters/11.txt
  - https://hosts.oisd.nl
- Go back to the "Home" section check for updates and apply them.
- Close SoundCloud
- Go to Settings->Apps&Notifications->App Info->SoundCloud->Storage&Cache
- Clear the Cache & Clear the Storage
- Reboot your device (come check for new hosts version from time to time + reboot to apply)

## 14. Viper4Android FX

Viper4Android FX is the absolute best Audio FX / Equalizer you can get on your device. When you take the time to configure it right then you can make a €40 headset sound like a €100-150 one. It has been around for a long time and I really recommend it.

- You should just be able to download and install it through the Magisk Manager.
- However on Android 10 it isn't that easy. If you are on 10 I recommend following this guide, which did the trick for me: https://www.thecustomdroid.com/install-viper4android-android-10-guide/

## 15. AfWall+

This gives you a full fledged easy to configure FireWall making use of the powerful Linux iptables firewal right on your android device.

You can use it to explicitly deny or grant applications connection to the internet.

- You can download the latest version of AfWall+ through Aurora Store
- Enable it's exposed module and reboot
- Open the app, in the top right put it to block selected for the easiest setup
- Configure to your liking and don't forget to enable it with the 3 dots on the top tight

## 16. Facebook & Messenger

The big bad F is definitely equally terrible as Google, Amazon and all other big tech. And since with everyone on there we pretty much have no choice but to use it, but we have Open Source software watching our back with the great wrapper app Frost for Facebook. It even comes with adblocking.

- Open up Aurora Droid->Search for Frost for Facebook
- Download, install, login and enjoy.

## 17. Mobilism

Mobilism has started out as an application for books but nowadays it's the place to be if you want to *cough* test out paid applications.

Aptoide does kinda the same but I don't recommend it since it's gotten a reputation for also offering a lot of malicious apps and being malicious itself.

Download the latest version of Mobilism through your browser on Android:

mobilism.org/app

## 18. XPrivacyLua

By now I assume you know about app permissions you need to manage (allow block access to contacts, camera, microphone, location, …)

But some apps plainly crash or won't allow you to continue if you refuse them access to certain permissions. That is where XPrivacyLua comes in, this allows us to lie about / fake our stuff, which greatly enhances your privacy. Since even though some apps are refused to certain permissions, you'll see when you enable em in XPrivacyLua that they'll still be accessing them EVEN THOUGH YOU REFUSED EM. But now with this you should be safe most of the time, since they'll be feeded garbage instead of the real thing 😊

It'll take a while to set up since some apps do need access to certain parts of your phone to work, but it'll be worth it.

- You can download the latest FREE version of XPrivacyLua through Aurora Store or the ExPosed Manager
- Enable it in the EdXposed Manager under the Modules section and reboot.

After installation you'll need to setup the permissions for each app, I recommend the following default scheme, but think for yourself while applying, your phone app needs a mic so you can call and your camare needs your camera etc:

- Fake all the things!

- Besides the following:
    - Get Applications (You'll need it to share stuff cross apps, like a YouTube link to Messenger)
    - Get Sensors
    - Read account name (Otherwise often you'll have to re-login each time you open the app)
    - Read clipboard (If you want to be able to copy/paste text in and out of the app)
- If the app crashes, open XPrivacyLua, check the hour of when the app tried to access certain permissions, and try disabling the ones accessed one by one till it works.
- You might have to reboot to apply the changes to the permissions.

## 19. MiXplorer

MiXplorer is one of the best current day file explorers that will also allow you to use root access to browse and alter your system files.

You can download the latest FREE version of MiXplorer through your browser on Android:

https://labs.xda-developers.com/store/app/com.mixplorer

## 20. Titanium Backup

It's been around since forever, it might not look so fresh anymore, but it's still damn powerful.

Allows you to make extensive backups, allowing you to restore the app, the data of the app or both. But also has other features like freezing (= disabling and hiding the app, handy before uninstalling to check if it doesn't break anything) and uninstalling system apps.

You can download the latest FREE version of Titanium Backup through Aurora Store

## 21. GravityBox

GravityBox allows you to tweak and modify most of the system UI (User Interface) exactly to your liking. It has been around since Android 4 and get's a new version upon each Android Release.

- Normally you just can get it through the EdXposed Manager
- However for Android 10 it's still in Beta, but it can be downloaded from GitHub: https://github.com/GravityBox/GravityBox/tags

## 22. Substratum

**Substratum**: Awesome theming engine that works with overlays, however with Android 10 bringing dark theming out of the box it has fewer uses it still is a really neat app. Since you can fully dark-theme the system on pie or below with it + still are able to dark-theme apps that refuse to implement one, like DropBox. It isn't restricted to dark theming only but that's what I like. It is a little bit confusing to use initially but I'll show you the ropes.

- You can download the latest version of Substratum through Aurora Store
- It will require storage, display over other apps and root to work properly.

Substratum manages and applies the enabled overlays, but it doesn't come with any out of the box, you'll have to get them yourself in the form of an APK. I'll be using PitchBlack and Swift Black, you could get the PAID versions through Aurora store. Now how to enable overlays:

- Open Substratum -> Select themes -> Select the theme you want to try overlays from
- Select your Android version (LineageOS is AOSP based)
- Select/configure the overlays you want
- Once you have all you want tap the round button with the overlay icon in the right bottom
- Select Build & Install then reboot your device
- Open Substratum again -> Select Manager (Here you can manage all installed overlays, installed isn't the same as enabled)
- In the top right to the left of the triple dots you can toggle between which overlays the manager will show you (All, Enabled, Disabled)
- Select the overlays you would like to enable
- Once you have all you want tap the round button with the overlay icon in the right bottom
- Select Enable Selected
- Open the apps and enjoy the theme, if it doesn't work try another reboot or another overlay pack.

## 23. Jasi Patcher

This will allow us to hack in-app purchases without the need of the Play Store on our device, like Lucky Patcher needs. Thus allowing us to use it in combination with MicroG.

- Download the apk here: https://jaspreetsingh.store/jasi-patcher/
- Open it up, go to Settings:
  - Disable Animation
  - Set Font to System Default
  - Enable AutoUpdate
- Go to Donation:
  - Enable Premium or donate if you're feeling frisky.
- Go to Dashboard->Emulation:
  - Enable the Allow display over other apps setting (do it manually if it doesn't pup up)
  - Enable License Emulation
  - Enable Billing Emulation
- Open up EdXposed Manager:
  - Enable the Jasi Patcher module and reboot

Now you can enable the Emulation for each app in JasiPatcher->Emulation->SELECT APPS (XPOSED). I would only enable it for the apps you want to use it with though.

## 24. CPU Info

This is a great good looking open source non spying app where you can find all the hardware specs of your phone in detail. Handy when you download .apk files through the browser since some apps are build towards a certain CPU architecture and with this you can check what you have on the fly.

Download the latest version of CPU Info through Aurora Droid or F-Droid

## 25. Split APKs Installer (SAI)

Some APK files come split up in multiple files nowadays. To be able to install these we'll need SAI.

Download the latest version of SAI through Aurora Store

Once installed open it up and give it root permissions.

Click Install APKs and select all the files needed then continue with installing them all at once. Sometimes you won't need all but will need to look at the CPU architecture with an app like CPU Info.

## 26. Keepass2Android & KeeWeb

These days I really recommend password manager that isn't owned by some capitalistic big tech company. For this I've been using KeeWeb in combination with Keepass2Android and I must say that they really work great.

KeeWeb is for your browser and Linux, Windows and Mac computers. While the other one has Android covered.

The benefit of these is that they allow you to save your password DB anywhere you want. You can store it locally, on your OwnCloud, Dropbox, GDrive, it's up to you. And you can move it whenever you want.

Also the ability too use authentication by a password + an encrypted key file that has to be on the device to use it adds greatly to the security of it. All while being free and open source.

One last thing, for autofill in your browser on a desktop you'll need a plugin but the official docs have a really clear guide for this: https://github.com/keeweb/keeweb/wiki/Browser%20AutoFill

Download the latest version of Keepass2Android through Aurora Store

## 27. WebKey

WebKey is a really good service to fully remotely control/stream your Android device, it does need root access but it's free for life for 1 device at a time.

Download the latest version of WebKey through Aurora Store

## 28. FairEmail

FairEmail might look a bit more dry then the bells n whistles you've been used too but this actually is a big one. It respects your privacy, while most other Email apps nowadays are heavily infested with trackers. Swapping to this will have a significant decrease in battery usage because of this. And it's open source!

Download the latest version of FairEmail through Aurora Droid

## 29. Customizing Termux

By default Termux looks like any plain black/white shell window, let's pimp it up though!

We'll be start out by doing a full Linux update, enabling the arrow keys, installing FISH, an awesome shell that has auto completion (By pressing TAB aka the two arrows between ESC and CTRL) and suggestions, color coded commands, etc. Then we'll add BobTheFish through OhMyFish, it's a Powerline style theme

for fish that will make your navigation bar look neat. A custom font will also be needed since the theme needs it to create icons.

Then we'll be setting up so fish will run on startup by default and slap some nice ASCII art we'll generate on it to finish it all off.

Lastly we'll install sudo, so we can access our terminal programs as root (**sudo su** to go into root mode)

- Open sd and type following commands:
  - apt update && yes | apt upgrade
  - yes | pkg install nano
  - nano ~/**.**termux/termux.properties
    - Enter following line in the file I suggest you copy/paste this piece:
    - extra-keys = [['ESC','/','-','HOME','UP','END','PGUP'],['TAB','CTRL','ALT','LEFT','DOWN','RIGHT','PGDN']]
    - CTRL, X, Y, Enter (To exit and save)
  - yes | pkg install fish
  - fish
  - curl -L https://get.oh-my.fish | fish
  - omf install bobthefish
  - pkg install git
  - git clone https://github.com/adi1090x/termux-style.git
  - cd termux-style && chmod +x install
  - ./install
- Now exit and restart Termux and type:
  - termux-style
  - F
- Now choose any font you like, they all have been Powerline-go patched, I'll be using Roboto.
- Generate a nice ASCII Art to your liking, I used this site: http://www.network-science.de/ascii/ _all fonts with your text_ is a handy option to look through em all.
- Create a text file ASCII-Art.txt and copy/paste the art into it.
- Save the txt file somewhere safe on your phone, I'll be storing mine in the Android folder (/storage/emulated/0/Android/ASCII-Art.txt)
- Now open up Termux again and type:
  - nano ~/**.**bashrc
    - Enter following 3 lines in the file I suggest you copy/paste them individually:
    - clear &&
    - cat /storage/emulated/0/Android/ASCII-Art.txt &&
    - fish
    - CTRL, X, Y, Enter (To exit and save)
  - pkg install ncurses-utils
  - git clone https://gitlab.com/st42/termux-sudo.git
  - cd termux-sudo
  - cat sudo > /data/data/com.termux/files/usr/bin/sudo
  - chmod 700 /data/data/com.termux/files/usr/bin/sudo

- sudo su
- nano ~/.bashrc
    - Add following line to the end of the file:
    - fish
    - CTRL, X, Y, Enter (To exit and save)
- curl -L https://get.oh-my.fish | fish
- omf install bobthefish

That's it, you should now have a beautiful customized shell! Only downside will be that you'll have to type exit 2 times now, once to exit from fish and once to exit from the normal shell.

Here are 2 extra fun packages you can install and try out:

- pkg install neofetch
- pkg install cmatrix

## 30. Conclusion

So that's my ultimate setup, completely free, fully UI customizable, hardware optimized, ad-free, private & premium! While keeping my system partition unmodified allowing me to easily flash system updates (As long as Magisk/EdXposed modules are disabled before updating!).

Hope you found this guide educational and have an awesome build as well by the end of this guide! Keep on modding!

## 31. Honorable Mentions

These are applications I still find worth mentioning but didn't include in this build.

- **Oreo Wifi Passwords:** Allows you to easily manage/share all the wifi passwords of the networks you ever connected to, however this feature comes out of the box with LineageOS 17.1
- **Tasker:** Write scripts to run on the background for automation purposes.
- **CrappaLinks:** Tries to resolve crappy links, browser to app, app to app etc.
- **QuickSwitch**: Switch your launcher (what renders your homescreen) to another one. For example I used this one too use OxygenOS's Launcher on LineageOS. But nowadays I'm more a fan of LOS's launcher.
- **BackgroundRestrictor**: Stop applications from running in the background to save battery, however in this build not needed since we have LineageOS's PrivacyGuard.
- **Amplify**: Wakelock and alarm blocker to extend battery life for Android 8 (Oreo) and below. Don't use on Pie and above. Android has had a big rewrite and is much more efficient in handeling these nowadays.
- **Greenify**: Kills apps running in background to save battery life. Don't use on Pie and above. Android has had a big rewrite and is much more efficient in handeling these nowadays. Also PrivacyGuard has got this covered.
- **RootCloak**: Should be obsolete since we have Magisk Hide nowadays for this, but you can give it a shot if Magisk Hide doesn't work for your application.
- **Boot Manager**: Don't think it still works on Pie and Privacy Guard also has this covered.
- **L Speed**: Huge library of hardware tweaks, comes with easy profiles for an optimized battery, balanced or performance experience. But I don't use it anymore. It feels like performance /

battery are better without it. And you really gotta ask yourself, who knows more about the hardware, you and the creator of L Speed, or the team of masters that made CyanogenMod and LineageOS.

- **Lucky Patcher**:Patcher like Jasi Patcher, but requires the Google Play Store to be installed. However I think their library of custom patches is bigger.